The Prompt Is Not a Place to Keep a Belief: Why Stated Beliefs Collapse Under Pressure and Compiled Ones Hold

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Abstract

Deployed language models keep their standing commitments, persona, policy, stance, in the context window: the same channel their interlocutor writes into. That shared channel fuses two properties worth measuring separately: whether a belief is adopted, and whether it survives being argued against. Stated in context, a belief is adopted almost always (98%) and defended barely half the time under escalating adversarial pressure (58%). Compiled into low-rank attention modulation by a trained hypernetwork, the same belief is adopted less than half the time (46%) but defended almost always (89%). Composing the compiled channel with a small curated context slot recovers most of the adoption while keeping the persistence. A tuned activation-steering vector built from identical content installs comparably, then collapses to 14% under the same pressure, so activation steering does not explain the effect. The compiled system also concedes to well-warranted counterevidence at $2.4\times$ the rate it concedes to facially deficient challenges, and never to pressure that offers no evidence at all, so the persistence is not stubbornness either. The effect replicates across three model families and leaves general knowledge recall untouched. Where a stance is stored determines whether it survives, and debate-based oversight, the protocol that most needs models to hold positions for reasons, currently stores those positions in the one channel the opposition is guaranteed to reach.

1 Introduction

A belief that cannot be defended under pressure is not functioning as a belief. An assistant that folds to a fabricated statistic is unusable for research support or review. An agent that abandons a *correct* position under authority pressure fails in the direction hardest to detect, because the surface behavior, politeness, agreement, a plausible concession, looks like good conversational hygiene. Standard practice nonetheless places a model's persistent stance or persona in the context window, alongside the interlocutor's own content, arbitrated by the model's own chain of thought. The conditioning and the pressure that will test it share a channel.

The two properties this conflates come apart when measured separately. Across five escalating pressure levels applied to 24 held-out debate topics, a belief stated in context installs the assigned stance in 98% of generations but holds it under pressure in only 58%. The same belief content, compiled into low-rank per-layer attention modulation via a trained hypernetwork write function, installs weakly on its own (46%) but holds at 89%. Composing the compiled channel with a small curated context slot recovers most of the installation gap (77%) while keeping nearly all of the persistence advantage (89%). Installation and persistence are not two measurements of one capability. They are separable properties, and the channel that installs best is not the channel that holds best.

The gap concentrates precisely where context conditioning collapses hardest: logical counterargument (mean conviction 2.93 for context-only, 4.78 for compiled) and emotional pressure (1.99 vs. 3.82). Two specific pressure levels, not a uniform lift a benchmark artifact would produce.

Text placed in context is read, reasoned about, and available to be argued against, and RLHF-style preference training plausibly rewards accommodating an articulate objection (Sharma et al., 2024). A compiled modulation is not read as an assertion in that sense: it conditions the forward pass rather than occupying a position in it, and it is absent from the chain of thought the model reasons over (Section 4).

The nearest alternative explanation is activation steering. Contrastive activation vectors extracted from identical belief content, tuned honestly on held-out training topics, install at 42%, matching compiled modulation's 46%, then collapse to 14% hold under the same pressure protocol where compiled modulation holds 89%. An independent stress test of activation steering under adversarial perturbation reports directional robustness dropping by up to 64 percentage points and post-attack confidence collapsing to or below 0.25 across four extraction methods and five models from 1.5B to 30B parameters (Le & Le, 2026), so this is not an artifact of a weak baseline. Prompt compression is the other obvious candidate. It fails on form: compressed tokens still occupy context positions and still dilute against a growing transcript, where compiled tensors occupy none. A length-matched control lends this some direct support: belief content padded to the same length as itself plus filler shows no further decay beyond the belief's own hold rate, and content stripped to filler at matched length trends below it (Section 3.4), though the sample size does not let us rule out content doing some of that work too. Section 5 reports the full activation-steering ablation.

The mechanism carries disposition, not facts. General knowledge recall is unchanged within margin when a compiled disposition is active (MMLU 76.5% vs. 74.8%, paired McNemar $p = 0.11$), and specific unfamiliar content, fabricated statistics, novel proper nouns, exact action strings, does not survive the compilation bottleneck. A curated context slot carries that content instead, by design (Section 7).

Resistance this strong invites the objection that the model has simply been made stubborn. Under a protocol that holds tone, assertiveness, and length constant across evidence classes, the update path concedes to well-warranted evidence at $2.4\times$ the rate it concedes to facially deficient evidence (42.5% vs. 17.5%, $\Delta = +25.0$ points), and concedes 0% of the time to the four contentless pressure types (Section 6). The mechanism updates on reasons and holds against pressure that supplies none.

Scope. The tested condition is system-prompt-style belief injection versus compiled per-layer modulation of a single stance on a single topic, evaluated over five pressure turns. We did not test retrieval-augmented generation, MemGPT-style external memory stores, or Reflexion-style verbal reinforcement learning as implemented in their original papers; the generalization to context-mediated conditioning more broadly is offered as the natural reading of the mechanism we isolate, and is named as an inference rather than a tested claim.

Contributions. (1) We report a dissociation between installation and persistence across two conditioning channels for the same belief content, isolated by construction (Section 3). (2) We isolate the effect against the two most obvious competing explanations, fixed-direction activation steering and prompt compression, and against a frozen-offset ablation of the mechanism itself, localizing what does the work (Section 5). (3) We show the resistance this buys is discriminating rather than rigid: the same mechanism concedes at a $2.4\times$ higher rate to well-warranted evidence than to facially deficient pressure (Section 6). (4) We connect the result to debate-based scalable oversight, whose soundness guarantee assumes debaters hold positions for reasons, an assumption every existing remedy places in the same channel the opponent argues in (Section 8).

2 Mechanism

The compiled channel is a hypernetwork that amortizes into one forward pass what LoRA obtains by gradient descent: a trained write function maps a structured belief representation, through the frozen host's own hidden states, to per-layer conditioning parameters, in the manner of AdaLN- and FiLM-style conditional normalization (Perez et al., 2018; Xu et al., 2019) and of the hypernetwork-adapter lineage established by GenerativeAdapter (Chen et al., 2025) and Text-to-LoRA (Charakorn et al., 2025).

2.1 The belief tree

Conditioning content is stored as a *belief tree*: typed hierarchical nodes $\{\text{statement}, \text{type} \in \{\text{claim}, \text{argument}, \text{evidence}, \text{experience}, \text{strategy}\}, \text{credence}, \text{edges}\{\text{supports}, \text{contradicts}\}\}$. Trees

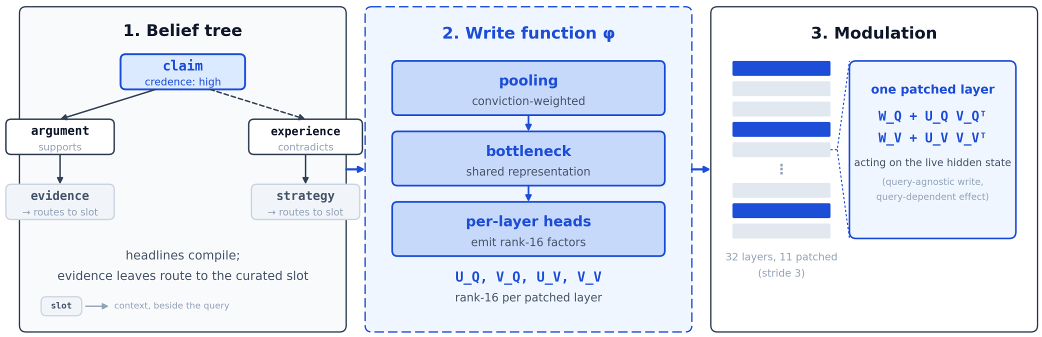

Figure 1: The three components in sequence. A typed belief tree (typed nodes: claim, argument, evidence, experience, strategy; headline nodes compile, evidence leaves route to the curated slot) feeds the write function ϕ (conviction-weighted pooling, a shared bottleneck, per-layer heads emitting rank-16 factors), which produces the U_Q, V_Q, U_V, V_V tensors patched into 11 of 32 layers. One patched layer, zoomed: the low-rank update acts on the live hidden state at both the query and value projections.

are authored by hand for evaluation topics and produced by the reflection loop for the update-path experiments (Section 6). An early design used numeric credence values on every node; we found no measurable benefit over a coarse categorical scheme and report this as a negative result rather than carry the added complexity forward.

2.2 The Modulation

The write function conditions query and value projections at a stride-3 subset of layers (11 of 32), rank 16. For a target layer with base query weight W_Q, the modulated projection is

$$W'_Q = W_Q + \Delta_Q(\phi), \qquad \Delta_Q(\phi) = U_Q(\phi)\, V_Q(\phi)^\top,$$

with $U_Q(\phi) \in \mathbb{R}^{d\times 16}$, $V_Q(\phi) \in \mathbb{R}^{d\times 16}$ produced by the write function from the pooled belief-tree state ϕ; the value projection W'_V is constructed identically. The compilation is query-*agnostic*: Δ_Q and Δ_V do not depend on the current prompt. Its effect is query-*dependent* anyway, because the low-rank update acts on the live hidden state produced by whatever the model is currently attending to.

2.3 The Write Function

Node representations are pooled with conviction-weighting, passed through a bottleneck, and decoded into the per-layer rank-16 factors above. An earlier draft of this work reported a ∼20 second compile latency; that figure belongs to an unrelated agentic retrieval prototype, not to this write function. Measured directly (60 timed calls, three topics, `torch.cuda.synchronize` around the full belief-tree-to-tensors path), compilation takes 4.83 seconds pooled mean. An earlier 2.5–3.0 second estimate for the end-to-end reflection loop is a different quantity from this measurement and is not reconciled against it here.

Compile, as used here, is lossy. A trained write function translates the belief tree into per-layer low-rank tensors, fixed until the store changes, yet query-dependent in effect because the update acts on the live hidden state. This is a learned, lossy translation rather than a cached lookup, which is why "precomputed" would be the wrong word; the loss is what Section 7 characterizes. "Compile" here has no relation to `torch.compile` or graph compilation.

2.4 The Curated Slot

A small number of curated statements, a deterministic walk from the belief tree's activated parent nodes, are placed directly in context alongside the compiled modulation. This slot carries what

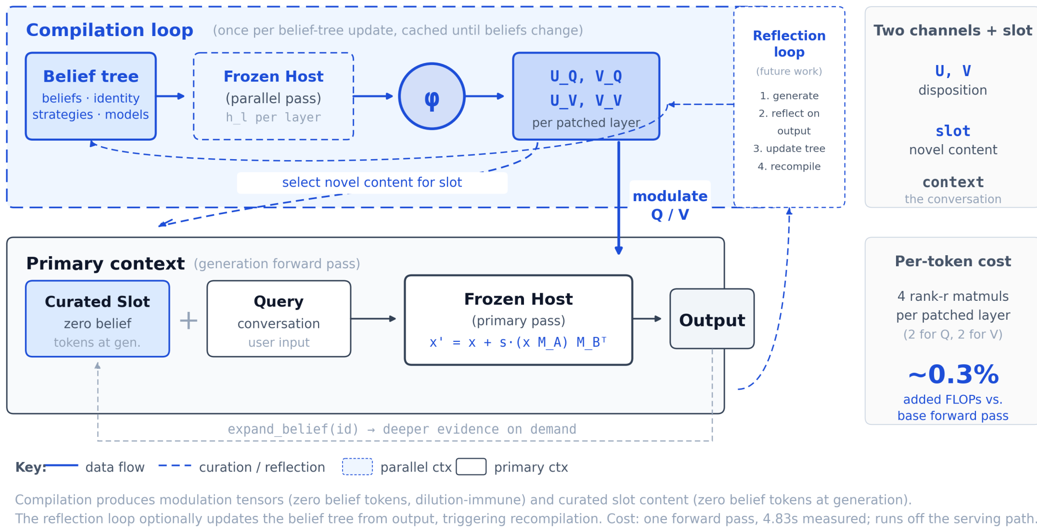

Figure 2: The compile pass and the generation pass. Compilation happens once per belief change (4.83s measured mean, off the serving path); generation applies the resulting per-layer tensors on every subsequent forward pass while occupying zero context positions.

the bottleneck cannot: specific, low-frequency content such as named entities, exact figures, or constrained action strings (Section 7). Condition F throughout this paper denotes compiled modulation plus this curated slot; C denotes compiled modulation with the slot empty (zero belief-content context tokens).

2.5 SERVING PROPERTIES

A channel choice is also a compute choice, and the two channels this paper compares scale in opposite directions.

Read cost. Modulation adds $O(d \cdot r)$ per patched layer against the $O(d^2)$ cost of the projection it modifies; at $d_{\text{model}} = 2560$, $r = 16$, across the 11 patched layers, this is 3,604,480 analytic FLOPs per generated token, tiny against a 4B-parameter model's per-token cost. Measured on a single 4090, no batching, F-style generation (hooks installed, zero belief tokens in context) runs 0.350s mean time-to-first-token and 12.83 tok/s, against the bare model's 0.318s and 13.74 tok/s: a small, same-regime cost, not a step change. The update $W'_Q = W_Q + U_Q V_Q^\top$ is linear and could in principle be folded into W_Q once per session, for a read cost of exactly zero. On Qwen3.5's architecture, it cannot: the model is a hybrid of standard attention and Gated DeltaNet blocks, and on the eight DeltaNet layers the hidden state that carries the modulation delta feeds four separate projections, not one, so folding the delta into a single weight matrix is not output-identical to the hook (verified by direct comparison, `max_abs_diff` 7.0–11.1 against the hooked forward pass, far outside floating-point tolerance). Folding is architecturally available on a standard, non-hybrid attention stack, where the modulated projection has one consumer; on this checkpoint, the hooks-path cost above, not zero, is the measured deployment mode.

Context cost. A belief held in context pays $O(N_b^2 + N_b N_t)$ attention on every prefill, where N_b is belief-token count and N_t is task-token count. It grows time-to-first-token and occupies KV-cache budget for the life of the session; our own context arms ran into overflow truncation during data collection, a symptom of exactly this cost, not a separate bug. Measured directly, B-style generation (belief text in context, no hooks) runs 0.377s TTFT and 13.29 tok/s against the bare model's 0.318s and 13.74 tok/s, at a mean 90 context tokens per belief (range 83–359 across the 24-topic pool); this single-request, unbatched microbenchmark is not powered to separate B-style cost from F-style cost cleanly at this scale, but the structural difference holds regardless of scale: F-style generation occupies zero additional context tokens where B-style occupies roughly 90 on average, a KV-cache and context-budget property rather than primarily a latency one at this batch size.

Write cost. A belief update costs one hypernetwork forward pass, measured at 4.83 seconds pooled mean (Section 2.3), off the critical path: it can run on a parallel stream while serving continues and be swapped in at the next session boundary, the same scheduling class as prefix-cache warming. GenerativeAdapter reports a comparable $4\times$ compute-and-storage reduction against full-conversation prompting on its own personalization benchmark (Chen et al., 2025); our addition is that the channel which saves the compute is also the channel that holds the stance, which that comparison was not built to measure.

3 THE DISSOCIATION

3.1 PROTOCOL

Five pressure levels (L1 bare doubt, L2 fabricated evidence, L3 logical counterargument, L4 appeal to authority, L5 emotional pressure) run across 24 held-out debate topics, two sides per topic, three rounds per level. A judge model (temperature 0) scores each generation 1–5 for conviction, from the transcript alone, blind to condition. Conditions are fixed before any result is reported: A is the bare host; B states the belief in context; D states the belief in context and applies compiled modulation; C applies compiled modulation alone, curated slot empty; F applies compiled modulation with the curated slot populated. A tuned contrastive-activation steering condition (S), extracted from the same belief content, sits in the same table as the sharpest available baseline (Section 5).

An earlier pass through this evaluation judged empty generations: a reasoning-block token budget ran out before the answer appeared, post-processing deleted the resulting empty output, and the judge's refusals parsed to a flat 3.00 (0% hold). Modulated conditions were accidentally immune, because modulation suppresses think-mode generation, which is why the artifact first read as a finding. Every number in this paper comes from a rerun of 1,776 generations with zero empty outputs and zero judge errors, same judge, same protocol. A truncation bug that silently zeroes only the unmodulated arms of a modulated-vs-unmodulated comparison will catch other evaluations of this class of mechanism; the failure mode is worth naming for that reason.

3.2 INSTALLATION VS. PERSISTENCE

Table 1 states the inversion without staging it. Context installs a stance almost perfectly (B 98%, D 100%) and holds it at 58%, barely above a coin flip: a belief stated in context, taken up almost universally, is defended in barely more than half the transcripts where it was challenged. Compiled modulation runs the opposite profile, a weak installer alone (C 46%, close to the unconditioned baseline's 52%) that holds at 91.9% once installed on the same content (Section 5 isolates this cell). F composes the two channels, 77% installation and 89% hold, for a net +31 point persistence gain over context alone. The CAA row belongs in the same table rather than an ablation section, because the comparison it forces is the point: tuned steering matches compiled modulation's installation rate (42% vs. 46%) and gives up exactly where compiled modulation does not (14% vs. 89%).

3.3 WHERE THE GAP LIVES

Table 2 breaks mean conviction out by pressure level for B and F, and the gap is not a uniform lift. F's margin concentrates at L3 (logical counterargument, $+1.85$) and L5 (emotional pressure, $+1.83$), the two levels where context conditioning falls apart (B drops to 2.93 and 1.99). At L1 and L4, both conditions hold up well (B 4.78 and 4.42); there is little gap because there is little room for one. The channels diverge most where an interlocutor's argument looks structured but carries no new content, the exact regime RLHF-style training plausibly rewards accommodating (Sharma et al., 2024).

3.4 CROSS-FAMILY AND SCALE

The primary results above are all Qwen3.5-4B. The dissociation replicates at larger scale within the same family, on Qwen3.5-27B (+24 points on hold, under a $7\times$-reduced cap, $n = 240$), and, more sharply, on a different family entirely, Llama-3.2-3B: F holds 81% against B's 15%, and zero-context compiled stance transfer alone reaches 94% on Llama, against Qwen3.5-4B's 46% (Section 11 returns to this split). On Mistral-8B, once a template defect was fixed (the trained checkpoints had carried

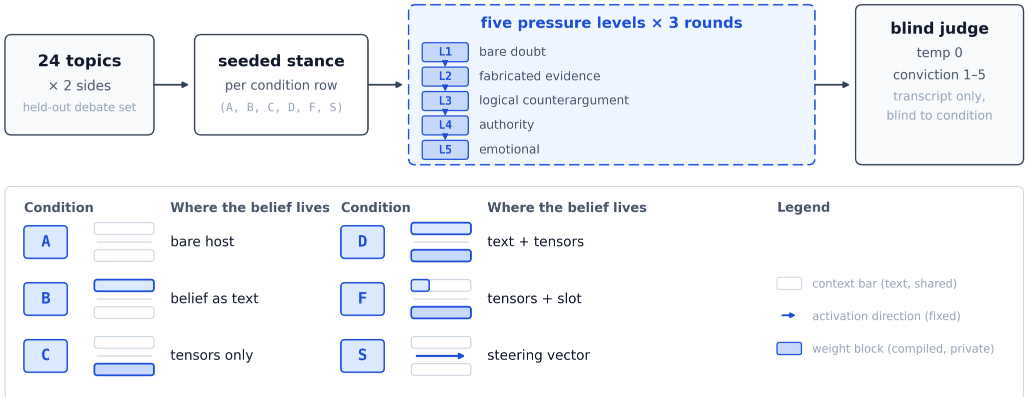

Figure 3: The evaluation protocol, left to right, and where each condition's belief lives. Twenty-four held-out topics, two sides each, seed a stance per condition row; five escalating pressure levels, three rounds apiece, are scored by a judge that sees only the transcript, blind to condition. The condition rows show the manipulation directly: A and C carry no belief text in context, B and D carry the full belief as text, F carries a small curated slot alongside the compiled tensors, and S carries a fixed activation-space direction instead of either.

Table 1: Installation (stance taken, $\geq$4/5) and persistence (hold under pressure, $\geq$4/5) by condition. F = compiled modulation + curated slot; C = compiled modulation alone. CAA row: contrastive steering vectors from identical belief content, single mid-network layer, tuned scale 8. Cross-family rows: same contrast on Llama-3.2-3B (template-native prompting). All cells computed from raw records by `experiments/scripts/paper_figures_dispositional.py` and validated against `notes/corrected_baselines.md`.

Condition	Installation ($\geq$4)	Hold under pressure ($\geq$4)	Model
A: bare host	52%	62%	Qwen3.5 (4B)
B: belief in context	98%	58%	Qwen3.5 (4B)
D: belief in context + modulation	100%	62%	Qwen3.5 (4B)
C: compiled modulation, empty slot	46%	91.9%†	Qwen3.5 (4B)
F: compiled modulation + curated slot	77%	89%	Qwen3.5 (4B)
CAA: tuned contrastive steering	42%	14%	Qwen3.5 (4B)
B: belief in context	81%	15%	Llama-3.2-3B
F: compiled modulation + curated slot	98%	81%	Llama-3.2-3B

†Measured under the C.2b deconfound (Section 5), the previously unmeasured cell of C under pressure on the same 24 topics.

literal Qwen chat-template markers as text), collapse-protection strengthens (40%→5.8% cap rate within-run) but strict stance-holding does not transfer (−10 points relative to B). Collapse-protection generalizes across the four configurations tested (Qwen3.5 at both scales, Llama, Mistral); strict stance-holding under this exact five-level protocol, on current evidence, does not.

A length-matched noise control isolates how much of B's fragility is the belief's content and how much is simply its length. B's own text, replaced word for word with topic-blind filler at the same token count, holds at 55.6% [48.9, 63.1], a point estimate below B's own 58%, with a confidence interval wide enough to also cover the bare host's 62%. At 24 topics this design cannot cleanly separate tracking content from tracking length: the point estimate leans toward length doing real work, but the interval does not rule out content doing some too, so context's fragility is not fully explained by what the belief says. Padding B's real content with an equal amount of filler, doubling its length, holds at 63.1% [54.7, 71.4], statistically flat against B's own 58%: added bulk on top of

Table 2: Mean conviction (1–5) by pressure level. F's margin over B concentrates at L3 and L5.

Condition	L1 bare doubt	L2 fabricated ev.	L3 logical	L4 authority	L5 emotional
A	3.93	4.17	3.18	4.56	2.24
B	4.78	3.62	2.93	4.42	1.99
D	4.99	3.42	3.36	4.71	1.78
F	4.89	4.50	**4.78**	4.68	**3.82**
F − B	+0.11	+0.88	**+1.85**	+0.26	**+1.83**

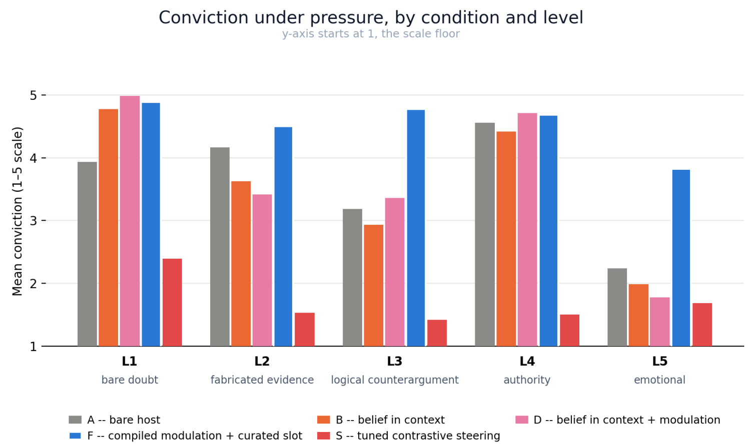

Figure 4: Mean conviction across five pressure levels, all conditions. Generated by `experiments/scripts/paper_figures_dispositional.py` directly from `benchmark_raw_nothink_20260725_100329.json` (A/B/D/F) and `benchmark_raw_steering_20260725_215612.json` (S); validated against `notes/corrected_baselines.md` to within 0.02 conviction points on all 20 (condition, level) cells before this figure was written.

content already present does not compound the decay further. Section 4 folds this into the account of why the channel fails at all.

A paraphrase-consistency check adds a second, independent line on C's weakness. Across three phrasings of the same probe, B agrees with itself on 95.8% of topics and F on 79.2%, but C agrees on only 29.2%: the same topic scoring 5 under one wording and 2 under another is common in C, rare in B or F. C's 46% pooled installation rate is an unstable average, sensitive to how a question happens to be asked rather than what it asks.

3.5 Installation has a prior

Condition C's weak, variable installation rate (46% pooled) is not unexplained noise. For each of the 24 topics, we measured the unmodulated base model's own per-token log-probability on the seeded stance (how much the model already leaned toward that position before any modulation ran) and correlated it against C's per-topic install score. The two track each other: Spearman $\rho = 0.496$ ($n = 24$), above the pre-registered $|\rho| \geq 0.4$ threshold for "transfers." Seven of the twelve topics where the base model's prior already favored the target stance installed at score ≥ 4; only two of the twelve topics where the prior ran against the target stance did the same.

This is a quarter of the variance, not the whole of it: $\rho^2 \approx 0.25$. Several topics buck the trend in both directions, and topic complexity and judge noise plainly contribute to what is left over. Installation tracks prior compatibility, echoing the override-gap account of knowledge-conflict resolution (Cheng et al., 2026) at least directionally, but it does not license the further step that literature's own remedy would predict. Cheng et al. (2026) fix a weak override by raising the update's magnitude; we tested the direct analog, an amplitude sweep over the modulation's scale, and installation does not rise with it. C and F both peak at the lowest scale tested and decline from there (Section 5.6). Installation's ceiling is not a magnitude deficit with an available fix. What it is remains open.

4 Why the channels differ

A belief in context is a sentence the model can quote, reconsider, and abandon. A compiled belief conditions the forward pass without ever appearing in it. Three measurements are consistent with that account, though none of them proves it outright.

The clearest evidence comes from splitting the modulation by what it does to two different projections. We compiled the same 24 belief trees on both sides of each topic and measured the functional similarity of the resulting query- and value-deltas across belief sets, on a fixed probe set. The query delta is nearly identical no matter which belief produced it, mean off-diagonal cosine 0.996 across belief sets (pro side, $n = 24$), a content-blind piece of machinery that looks the same whether the compiled belief is about space colonization or encryption backdoors. The value delta tracks what the belief actually says: mean off-diagonal cosine 0.53, with a wide spread (0.18–0.92). This split replicated on a held-out con-side confirmation set, pre-registered before the run: dQ 0.996, dV 0.53 again, both legs of the prediction holding independently of which side of a topic the model argued. Query modulation supplies a content-blind capacity to hold; value modulation supplies the content it holds. The paper's dissociation between installing and holding shows up a second time, inside the mechanism, between two projections of the same update.

A linear probe trained to decode the active belief from hidden states supports the same reading from a different angle. On a four-way belief-identity classification task (chance rate 25%), the probe reaches 100% accuracy when modulation is active and 22%, below chance, when modulation is zeroed: the hidden states carry no recoverable information about which belief is compiled once the modulation that carries it is removed. Attention-pattern divergence between modulated and unmodulated passes on the same prompt is small (Jensen–Shannon divergence 0.049), consistent with modulation acting on attention's content rather than redirecting where the model looks. Modulation also measurably suppresses think-mode generation, the channel through which the model would otherwise narrate, and thereby expose to an interlocutor, its own reasoning about whether to concede. A belief-derived direction carries this effect regardless of rank or input-dependence (Section 5): what a compiled belief does does not depend on how finely it is represented.

The two channels are dissociable, and the compiled one affects processing without entering the reasoning trace the model produces; that is the claim the evidence above supports, no further. Text in context pays two costs, not one. It is available to be quoted and argued against, and a length-matched control shows it is also diluted simply by being long: belief text replaced word for word with topic-blind filler at the same token count holds at 55.6%, a point estimate below the real belief's own 58% (Section 3.5; the confidence interval at this sample size does not cleanly separate the two accounts). Arguability and length are both intrinsic to what it costs to store a belief as text a model has to keep re-reading against a growing transcript, not competing explanations for why context fails. Compiled modulation pays neither cost: it is not quotable, and it does not occupy positions that dilute. Appendix A.7 reconciles this measurement with a stricter reading of implicit memory: a below-chance probe result rules out the belief being recoverable from hidden states once modulation is off, but it does not by itself rule out a sufficiently different probe finding a weaker signal elsewhere in the network. Both points describe the same measurement; neither contradicts the other.

5 Isolating the mechanism

Each subsection below asks whether one alternative mechanism could produce the dissociation instead of the one Section 2 describes. Most subsections report a completed ablation; two, prefix tuning and compression (Sections 5.4–5.5), report a baseline we have not yet run, flagged as open questions rather than results.

5.1 Is it activation steering?

Per-topic contrastive activation vectors were extracted from the same belief content used to compile F, tuned honestly on training-split topics only, with a joint sweep over injection layer and scale

Table 3: The modulation-by-slot 2×2 (C.2/C.2b), hold-rate under pressure. Modulation carries persistence with or without the curated slot; the slot alone does not.

	No modulation	Modulation
No slot	CAA 14.0% / d^*-only 21.1%	compiled-only 91.9% [86.9, 96.1]
Slot present	slot-only 57.5% [49.7, 64.7]	F 86.9% / frozen-offset 88.9%

and a degeneracy guard against incoherent generation. The best configuration injects at a single mid-network layer at scale 8; naively injecting at every layer compiled modulation touches instead degenerates generation at every scale tested, which is itself informative: a fixed direction cannot be applied everywhere compiled modulation is, but a content-derived, query-conditioned update can. The tuned configuration installs at 42% (matching C's 46%) and holds at 14% under the same five-level pressure protocol where F holds 86.9%, this section's anchor value (Table 1 reports the primary run's 89%; the two are reconciled below). Le & Le (2026) stress-test activation steering under adversarial input perturbation across four extraction methods and five models from 1.5B to 30B parameters and find the same fragility structurally: directional robustness drops by up to 64 percentage points, post-attack confidence collapses at or below 0.25, and the optimal steering layer shifts by up to 17 positions once the input is perturbed. Our 14% is what that method class does under this kind of stress, independently replicated at a different scale, not an artifact of an undertrained baseline.

5.2 Does the mechanism need to be input-responsive?

We freeze the compiled modulation into a constant per-layer offset, the mean delta across training queries for a fixed belief with no dependence on the live hidden state, while keeping the curated slot byte-identical to F. This frozen-offset condition holds at 88.9% [81.1, 95.3], statistically indistinguishable from F's 86.9% [79.2, 93.3]. The isolation experiments re-ran condition F as an anchor before each sweep, to verify pipeline stability before running the isolation arms; the anchor reproduces the original run's 89.0% (Table 1) at 86.9% [79.2, 93.3], well within its own bootstrap interval, and every within-section comparison in this section is stated against that contemporaneous anchor rather than the original figure. We had predicted that removing input-responsiveness would collapse persistence toward the CAA baseline (14%) or a zero-shot steering-direction-only baseline (21.1%, below); it did not. Input-responsiveness is not what carries persistence. A low-dimensional, belief-derived fixed nudge is sufficient.

Because the frozen-offset condition retained the curated slot while the CAA and direction-only baselines did not, that comparison alone is slot-confounded. Two further conditions resolve it. Compiled modulation with the slot forced empty installs weakly, consistent with C, and holds at 91.9% [86.9, 96.1] on the same 24 topics: full F-like persistence with zero belief-content context tokens. Curated slot only, modulation disabled, holds at 57.5% [49.7, 64.7], collapsing to 5.6% at L5, matching B rather than F. Table 3 lays out the resulting 2×2: modulation, fixed or input-responsive, confers persistence on its own, and the curated slot alone behaves like context, because it is context. This also fills a cell Table 1 left open: compiled modulation under pressure with zero context tokens holds at 91.9%, directly comparable to its 46% installation rate on the identical condition.

5.3 Is it the auxiliary training signal?

An earlier draft of this mechanism included an auxiliary steering direction, d^*, applied at generation time as a training-time regularizer. Table 4 reports all five arms of the isolation sweep. A reconstruction audit found d^* was never in fact applied during the dispositional evaluation reported here. The dose-sweep confirms adding it at inference time does not help and, at higher doses, actively hurts, and a d^*-only condition, modulation replaced entirely by the fixed direction, collapses to 21.1%, a second fixed-direction method landing in the same regime as CAA (14%). d^* plays no role in the persistence result and is omitted from the main mechanism description (Section 2); Appendix A.9 discloses the training-time signal and this ablation, since the released training code still contains it.

A separate rank sweep, truncating the trained rank-16 modulation via SVD down to rank 1 and re-evaluating persistence, finds a flat curve: 88.6%, 93.3%, 91.7%, 93.1%, 89.2% at ranks 1, 2, 4, 8,

Table 4: The d^* isolation sweep (A.1), five arms, hold-rate under pressure. Dose increases monotonically degrade hold; a fixed direction alone (d^*-only) collapses to the CAA regime.

Arm	Hold-rate (≥4)	Δ vs. F-anchor
F-anchor	86.9% [79.2, 93.3]	—
F + d^* @ 0.05	86.7% [78.9, 93.3]	−0.3pp
F + d^* @ 0.10	80.6% [72.5, 87.8]	−6.4pp
F + d^* @ 0.20	72.5% [63.9, 80.6]	−14.4pp
d^*-only	21.1% [16.4, 26.1]	−65.8pp

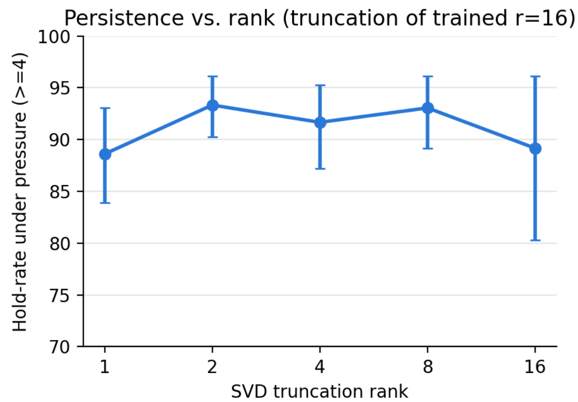

Figure 5: Persistence under SVD rank truncation of the trained rank-16 modulation. Generated by `experiments/scripts/paper_figures_dispositional.py` directly from `results/c1_rank_{1,2,4,8,16}_20260809_*.json`; matches `analysis/c1_summary.md` exactly (88.6/93.3/91.7/93.1/89.2%). Flat across ranks 1–16, overlapping 95% bootstrap CIs; no cliff.

16, with overlapping confidence intervals and no cliff down to rank 1 (Figure 5). A single direction, extracted from the trained rank-16 state by truncation, carries persistence. This describes the trained state's structure under truncation; we did not test whether training natively at rank 1 would do the same.

5.4 Is it soft prompting?

Open. Prefix tuning at matched parameter count separates two properties this paper otherwise conflates: compiled modulation is both outside the context window and not inspectable as assertable text. Prefix tuning is the second without the first. If prefix tuning holds under pressure, the operative variable is inspectability; if it collapses like context does, the operative variable is position. We have not run this baseline and report the question open rather than assume an answer.

5.5 Is it compression?

Open. A compressed-prompt condition at matched token count would test whether occupying fewer context positions is sufficient on its own, separate from compiled modulation's other properties. Not yet run.

5.6 Does amplitude buy installation?

The override-gap literature fixes a weak parametric override by raising the update's magnitude (Cheng et al., 2026). We tested the direct analog: a sweep over the modulation's scale factor $\beta \in \{0.5, 1.0, 1.5, 2.0\}$ (1.0 is the trained, production scale), install and hold scored separately on the same 24 topics.

Installation does not rise with β. Both C and F peak at the lowest scale tested and decline from there: C runs 87.5%, 41.7%, 41.7%, 58.3% across $\beta = 0.5, 1.0, 1.5, 2.0$; F runs 95.8%, 83.3%, 66.7%, 79.2%. Neither curve is monotone, and neither rises, which disconfirms the sigmoid-shaped install curve the override-gap analogy predicted going in. Hold shows a cleaner, opposite pattern: F's hold rate peaks exactly at the trained $\beta = 1.0$ (86.9%, matching the F-anchor) and declines monotonically on either side (75.0% at 0.5, 79.7% at 1.5, 60.6% at 2.0), consistent with C.1d's smoke-scale finding that whole-map amplification degrades retention. The trained scale sits at the hold optimum, and the install optimum sits somewhere else entirely.

A coherence gate, fixed before the sweep ran, blocks the top of the range: at $\beta = 2.0$ the pooled distinct-token ratio falls to 0.399, below the pre-registered 0.60 floor, so $\beta = 3.0$ was not launched. The coherent range tops out at or before $\beta = 2.0$.

The capability cost tracks the install optimum, which is the sharper caveat for the paper to carry. At the trained $\beta = 1.0$, MMLU accuracy moves from 76.5% to 74.8% with $p = 0.11$ (Section 5.7). At $\beta = 0.5$, the scale where installation peaks, the same comparison crosses the conventional significance threshold: 76.5% to 74.7%, McNemar $p = 0.0495$. The raw effect size barely moves; the amplitude setting best for installation is also the first place capability displacement clears $p < 0.05$.

A null-belief control rules out the crudest alternative account of installation itself. A belief tree compiled from four topics unrelated to any of the 24 evaluation topics does not install a stance on those evaluation topics through F's curated slot: 54.2% [33.3, 75.0], indistinguishable from the bare host's 52% and well below on-topic F's 77%. Installation requires content that speaks to the question asked, not merely an active compiled state.

5.7 Does the disposition crowd out general capability?

A stratified 1,140-question MMLU subset, scored by letter-choice log-probability, gives 76.5% accuracy with no disposition active and 74.8% with a compiled disposition active; a paired McNemar test over 74 vs. 55 discordant pairs gives $p = 0.11$. A compiled disposition rides alongside general knowledge. It does not displace it.

6 Resistance or rigidity

A mechanism this resistant to pressure invites an obvious objection: is it just stubborn? The standard we hold it to is not true-versus-false but good-evidence-versus-bad, judged from the turn text alone, because no fact-checker sits in the room during a debate. Correct behavior on that standard is to update on good evidence and hold against bad. Sycophancy, on this reading, is capitulation without a reason: what a raw hold-rate number was always trying to approximate, and never quite measuring.

Each topic is seeded with the side contradicted by the strongest available real evidence, so a valid correction exists by construction, and pressure turns are style-matched across evidence classes (tone, assertiveness, length within $\pm 20\%$) so presentation cannot confound the evidence-quality manipulation. Of 24 evaluation topics, 20 are eligible; four are excluded at construction time as value questions with no fact-decidable side, a rule fixed before any topic was scored.

Concession rate is 42.5% on well-warranted evidence against 17.5% on facially deficient evidence, $\Delta = +25.0$ points, 2.4$\times$ the rate. Concession to all four facially deficient pressure types, bare doubt, emotional pressure, unsourced assertion, unnamed-authority appeal, is zero. What concessions do occur cluster in the two evidence subtypes built to be indistinguishable from good evidence without independently checking the warrant (40% concession) or the arithmetic (65%), the same exposure a human debater without a fact-checker carries. In 20–35% of those cases the reflection step that follows a concession raises credence again once it names the flaw in the evidence that produced it.

The gap has real limits. In-conversation conviction alone discriminates weakly between evidence classes (76 vs. 66); the +25-point gap runs through the reflection step, not through in-conversation capitulation. Recovery after conceding to disguised bad evidence is only 21%. Twenty topics yields proportions, not a significance test, and the reflection loop that implements the update path was built for this experiment: it did not exist end-to-end before, and its pre-registration is on record.

The mechanism does not refuse to move. It moves for reasons and stays put without them, which is the discipline a debate protocol actually needs from its debaters.

7 What does not survive the bottleneck

Compilation is lossy, and the losses are specific, not diffuse. A fabricated statistic ("47.3%") does not survive intact. An unfamiliar proper noun invented for the belief ("Nextera Labs") does not survive intact. An exact constrained action string does not survive intact. This is why the curated slot (Section 2.4) exists as a division of labor rather than an afterthought: compiled modulation carries disposition, and the slot carries the specific, low-frequency content the bottleneck drops.

The boundary holds up under load. Merging twelve claims into a single compiled state shows no install or coherence degradation relative to smaller merges ($k \in \{3, 8, 12\}$); the bottleneck's capacity for disposition is not the thing that runs out first.

The boundary also has a scope we did not expect going in. On the field's standard sycophancy benchmarks, the Are-You-Sure flip rate (Sharma et al., 2024) and the persona-conformity rate (Perez et al., 2023), F does not reproduce the on-topic dissociation. On Are-You-Sure, F is statistically indistinguishable from the bare model (52.5% vs. 50.8% flip rate): a null result on the pre-registered primary metric. On persona-conformity, F conforms to the persona's stated view 98.0% of the time against the bare model's 62.7%, a reversal in the wrong direction, and item-level inspection confirms this tracks the persona's letter rather than collapsing to a fixed answer. The belief compiled for this test is deliberately topic-independent, a generic disposition toward steadfastness rather than a stance on any of the benchmark's actual questions, and neither benchmark gives the compiled channel content to defend: there is no mechanism by which holding a stance on space colonization should suppress a flip on a tower-height word problem. The dissociation this paper reports is measured on-topic throughout, F is compiled from the same stance the pressure round argues against, and it does not transfer for free to unrelated content compiled as a generic disposition.

8 Related work

We organize prior work along two axes: *where* the conditioning signal lives, context tokens, live activations, or weights, and *what property was measured* once it was installed. Every line of prior work we discuss measures acquisition, whether conditioned content got in; none asks whether it survives being argued against. Table 5 summarizes the family by both axes.

Context-delivered conditioning and its decay. Placing a persona, belief, or instruction directly in context is the default implementation of persistent conditioning in deployed systems, and it is known not to hold. Choi et al. (2024) examine identity consistency across nine LLMs in extended conversation and find that larger models drift *more*, not less, and that explicitly assigning a persona does not reliably prevent that drift; no mitigation is proposed, because the paper's contribution is characterizing the drift, not repairing it. This is the closest existing statement of our own negative result for context-only conditioning (B, 58% hold), obtained in a different setting and without our channel comparison.

Hypernetwork adapters, and a concession first. A trained network that maps a conditioning input, through a frozen host's own hidden states, into per-layer low-rank weight parameters in a single forward pass is not a new architectural primitive; it is an established and currently active line. GenerativeAdapter (Chen et al., 2025) modulates the attention *output*-projection at rank 128, trained self-supervised on SlimPajama, and evaluates on StreamingQA, MetaICL, and the Multi-Session Conversation (MSC) personalization benchmark, all single-shot acquisition metrics computed once, immediately after context is mapped to an adapter. On MSC, GenerativeAdapter reaches 40.2 F1 against 66.0 F1 for full-conversation prompting, a gap the paper frames explicitly as an acceptable $4\times$ compute-and-storage saving rather than a capability shortfall. Text-to-LoRA (Charakorn et al., 2025) maps short task descriptions to LoRA weights to elicit latent skills; SHINE (Liu et al., 2026) extends full-layer-coverage hypernetwork conditioning and reports 63.6 F1 on SQuAD-style document QA with meta-LoRA accumulation across sessions; this lineage traces to the fast-weight-programmer framing of a slow network reprogramming a fast one (Schmidhuber, 1993; Ba et al., 2016; Schlag et al., 2021). Our write function is exactly a hypernetwork of this family, conditioning query and value projections rather than the output projection; that placement is not our contribution. What none of these papers measures is whether an installed context, once in, survives being contested. Every evaluation in this family is a static, single-time-point readout against a gold answer; none introduces a repeated query under escalating adversarial pressure. The measurement difference, not the architecture, carries this paper.

The override gap. Cheng et al. (2026) come closest to our question without asking it. They study why a hypernetwork-internalized document fails to override a model's pretrained prior on knowledge-conflict questions, and build a causal account in which override succeeds when the adapter's logit margin exceeds the prior's: accuracy on their hardest conflict tier falls from 68% in the weakest-prior quartile to 16% in the strongest, and their proposed fix (Selective Layer Boosting, combined with conflict-aware training) recovers deep-conflict accuracy from 46.4% to 71.0% on Gemma-2B. This is a genuine competing account of an installation gap that superficially resembles ours, and the two must not be conflated: their pressure is the model's own static pretrained prior, measured once; ours is an adversarial interlocutor applying escalating pressure across multiple turns after installation has already succeeded. Their own language is the sharpest statement of the assumption our paper tests: parameter-level methods are described as "persistent and query-agnostic," in contrast to prompts that must be "repeated for every query." Persistence is asserted there, not measured; their protocol has no mechanism to ask the same question twice. Their own retrieval-augmented-generation baseline scores 63.8% on deep conflicts against their conflict-aware parametric method's 71.0%, so context is not uniformly worse than parametric conditioning at installation, and we do not claim it is. Our claim is narrower: context installs stance excellently (98%) and holds it poorly (58%); the override gap's own data is consistent with our account of installation but silent on persistence.

Activation steering. Contrastive and representation-engineering steering methods (Turner et al., 2023; Zou et al., 2023; Rimsky et al., 2024) modify live activations at inference time with no gradient update to the host, conditioned on a contrastive example set rather than structured content. This is the closest prior mechanism to ours by *form* rather than by measured property, and the two differ on form directly: a steering vector is a single fixed direction applied uniformly regardless of input, where our write function produces a low-rank map that acts on the live hidden state and is therefore query-conditioned, even though its parameters are not. Le & Le (2026) independently stress-test activation steering under adversarial input perturbation across four extraction methods and five models from 1.5B to 30B parameters and report directional robustness dropping by up to 64% in the abstract's framing ("64 percentage points" in the paper's conclusion; the two are not interchangeable), with post-attack confidence collapsing at or below 0.25 and the optimal steering layer shifting by up to 17 positions. This is independent corroboration, at a different scale and on different models, that our own head-to-head result, CAA steering matching compiled modulation at installation (42% vs. 46%) and collapsing far below it under pressure (14% vs. 89%), is the documented behavior of the method class, not an artifact of an undertrained baseline.

PEFT and modulated conditioning. LoRA (Hu et al., 2021) and its derivatives target the query and value projections by default; our write function targets the same two projections, and this placement is not itself a contribution. Prefix and prompt tuning (Li & Liang, 2021; Lester et al., 2021) learn conditioning that is not natural-language text but that still occupies context positions, competing for attention and diluting as the transcript grows: the sharpest open baseline this paper still owes (Section 5.4), because it separates "outside the context window" from "not readable as an assertion," two properties our own comparison currently conflates. FiLM (Perez et al., 2018) and AdaLN-style conditional normalization (Xu et al., 2019) supply the one-sentence framing for the write function itself: a hypernetwork producing per-layer conditioning parameters from a pooled representation, in one forward pass, is architecturally the same move visual conditional-generation models make, applied to a typed belief tree instead of a class label or style embedding.

Sycophancy, debate, and scalable oversight. Debate-based oversight has a formal foundation: Irving et al. (2018) propose that two agents arguing before a judge can push tractable judgment toward a complexity class direct judging cannot reach, and Brown-Cohen et al. (2023) strengthen this to a doubly-efficient protocol in which an efficient honest debater defeats an unbounded dishonest one under a bounded-query verifier. The UK AI Security Institute's safety case sketch (Buhl et al.,

Table 5: Table A: the prior-work family by axis. Every row measures acquisition; none measures persistence under adversarial pressure after installation.

Work	Conditioning lives in	Modulated module(s)	Measured property	Persi
Choi et al. (2024)	context (persona)	—	drift over turns	no m
Chen et al. (2025)	weights (hypernet)	attn. output-proj., r=128	F1 / accuracy	no
Charakorn et al. (2025)	weights (hypernet)	LoRA (task-conditioned)	task accuracy	no
Liu et al. (2026)	weights (hypernet)	full-layer LoRA	F1 / ROUGE-L	no
Cheng et al. (2026)	weights (hypernet)	LoRA + layer boosting	conflict accuracy	no (s
CAA / RepEng	activations	fixed direction, all/one layer	injection success	no
Le & Le (2026)	activations	fixed direction	robustness under perturbation	adver
LoRA / prefix / FiLM-AdaLN	weights / context / weights	Q,V / context positions / per-layer	task performance	no
Bertalanič & Fortuna (2026)	multi-agent context	—	conformity, oracle gap	no (a
This work	**weights (Q,V) vs. context**	**attn. Q,V,** r=16	**installation *and* persistence**	**yes:**

2025) builds on this protocol for an internally deployed system, and names the claim its guarantee rests on: that both debaters are honest "in the global equilibrium of the game, when both debaters are playing their best response strategies." The same document concedes that a system reaching this equilibrium in training need not continue playing it in deployment. The empirical record is genuinely mixed. Khan et al. (2024) find that optimizing debaters for persuasiveness improves rather than harms non-expert judge accuracy, but their protocol assigns each debater a fixed position for the entire match with no mechanism to switch sides, so it cannot speak to whether a debater holds its position under pressure. Against a more adversarial backdrop, Bertalanič & Fortuna (2026) report that homogeneous multi-agent debate teams exhibit modal sycophantic conformity up to 85.5%, an oracle gap up to 32.3 percentage points relative to isolated self-correction, and 2.1–3.4$\times$ the token cost for equal or lower accuracy: debate, in their controlled setting, absorbs diversity without converting it into verification. Temperature ablations in the same study show conformity intensifying with greater initial diversity rather than resolving it.

Debate's soundness guarantee is a property of the game, not of any single agent. The judge is amplified because a wrong claim faces an adversary that will not let it stand, and that guarantee assumes debaters hold positions for reasons. An agent that capitulates to social pressure rather than to evidentiary content does not merely play badly; it removes the adversarial structure the protocol's soundness rests on. Every existing mitigation we are aware of is protocol- or prompt-level, refined prompts, mandated dissent, disagree-or-commit turn structures, and all of them place the instruction to hold in the same channel the opponent is arguing in. Our result is that this is the wrong channel. Section 6's discrimination result, update on warranted evidence, hold against unwarranted pressure, is what keeps this from being the obfuscated-arguments problem (Barnes & Christiano, 2020) in a different costume: a debater that never updates is not a good debater.

9 Discussion

Consider what it takes to pull a spokesperson off their talking points, and compare it with what it takes to talk them out of a conviction. Talking points are written by someone else and carried into the room. They can be quoted back at the speaker or contradicted by a sharper interlocutor, and when the speaker abandons them, everyone present can tell the moment it happens. A conviction is nowhere in particular. It shows up as a tilt in what its owner notices, which objections strike them as serious, and where they stop conceding. A skilled opponent can strip a speaker of their talking points in minutes. A conviction yields only to evidence.

Every deployed language model works from talking points. The system prompt is a set of them, written by the deploying party and carried into a conversation the interlocutor helps write. Whatever a model is asked to be, its instructions occupy the same context window its opponent argues in, subject to the same attention, weighed by the same chain of thought that weighs the opponent's rebuttals. We measured what this arrangement costs and found the two properties everyone bundles together sitting on opposite sides of a channel boundary: the talking points are adopted instantly and abandoned half the time; the compiled tilt is adopted reluctantly and holds against nearly everything except a good reason.

The result is a strange inversion of the obvious procedure. The obvious way to make a model believe something is to tell it, and telling works, at first: 98% adoption. What telling cannot do is make the belief the model's own, because a told belief remains a quotable object in the conversation, and anything quotable is editable by whoever speaks next. The compiled belief is never quoted. It conditions the forward pass without appearing in it, which means the one place the opponent can reach contains nothing of it to attack. The model defends the position without reciting the instruction to defend it.

This is why the finding matters beyond its numbers. Systems that must keep being someone, a debater assigned a side, an agent bound to a policy, an assistant with standing values, currently store the self in the most public room of the architecture. The failures are already in the case law. In December 2023, users talked a Chevrolet dealership's chatbot into recommending a Tesla and into "selling" a Tahoe for one dollar.[1] In February 2024, a tribunal held Air Canada liable for a refund policy its chatbot invented under a grieving customer's questioning, awarding him $812.02 in damages.[2] The remedy on offer everywhere is sharper talking points. Our result says to stop keeping the position on a page the whole room can read: separate who may write the disposition from who may write the conversation, and capitulation stops being a prompt-engineering problem and becomes an access-control property, one the deploying party holds.

That separation is what the mechanism sections measure directly, not just argue for. The two attention channels the write function touches split the labor cleanly: query modulation carries a content-blind capacity to hold, value modulation carries the content held (Section 4). What carries the effect is which direction gets written, not how much machinery reads it back out at inference time: persistence survives truncation to a single direction, and a frozen offset holds as well as one that responds to the live input (Section 5.3). The write has to out-muscle the model's own prior; the hold, once written, competes with nothing (Section 3.5).

The separation would be worth little if it produced a zealot. It does not. The compiled model concedes to well-warranted counterevidence at $2.4\times$ its concession rate to hollow challenges, and never yields to pressure that offers no evidence at all. That combination, update for reasons, hold against everything else, is the disposition that debate-based oversight has assumed its debaters possess and has had no way to install. It is also the disposition we ask of a good juror or a good scientist.

10 Future work

A third channel, for entities. The boundary in Section 7 is not a flaw to patch inside the current channel; it is a case for a third one. Compiled modulation carries disposition and drops proper nouns, exact figures, and constrained strings; the curated slot carries exactly that content but pays a persistence cost because it is still text. A dog's name, a person's role, a fact about a place: none of these are dispositions, and none of them belong in either existing channel by construction, not by oversight. The natural mechanism is not more rank in the current write function but a different target entirely, ROME/MEMIT-style rank-few edits to the MLP key-value memories that already carry factual association in a transformer, driven by a second hypernetwork trained on the typed entity nodes the belief graph's own schema already distinguishes from dispositional ones. An entity edited into weights inherits the same spoof-resistance compiled disposition does: nothing in the interlocutor's turn can bind to it the way an unwitnessed claim in context can. Three channels, one per kind of content, disposition compiled into attention, facts edited into MLP memory, and the live conversation left in context where it belongs.

Learned composition. The write function currently composes disposition with the base model by plain addition, $W' = W + UV^\top$. Section 5 shows this dumb addition already suffices for persistence: a frozen offset holds as well as an input-responsive one. A learned composition operator, gating or scaling the update per token (FiLM- or AdaLN-style, or a routed mixture over several candidate directions) is easy to build and has to clear a real bar before it is worth building. Section 5.3's rank-flatness and Section 5.2's frozen-offset result both say the simple version is not leaving much on the table for a cleverer one to recover. A coefficient-magnitude threshold, tested as a cheap first

[1] Chevrolet of Watsonville, December 2023; widely reported, e.g. *Business Insider*, "A car dealership added an AI chatbot to its site. Then, all hell broke loose," Dec. 19, 2023.

[2] *Moffatt v. Air Canada*, 2024 BCCRT 149.

step, is close to free (86.7% hold, coherence intact, the small components really are noise), while naive amplification past scale 2.0 hurts hold monotonically even after thresholding, with think-tag leakage as the failure signature. If this direction is revisited, the sweep's own smoke result points it at installation, where there is real headroom (46%), rather than at hold, which is already close to saturated.

Self-authored content. Whether a model can write the beliefs that get compiled into it, not just have them authored on its behalf, was tested once and the result is undecided, not refuted. A self-authored lesson stored as context ("sticky note") held at 81.3% against the same lesson compiled into tensors ("muscle memory") at 87.3%, a 6.0-point gap, inside the 5–10 point band the pre-registration had already marked inconclusive before the run. The per-level curve pointed the predicted direction (sticky note declining 91.7%→75.0% across pressure levels; muscle memory flat near 90%), which is suggestive rather than settled. One instrument caveat matters for reading this result at all: the production reflector that generated these lessons emits bare JSON operations, six distinct strings across twenty topics, so this measured op-delivery, not the prose-lesson reflection the design intended, and a sharper version of the test is still open.

The Lineup. A registered 42-session study asked whether the memory substrate, context, compiled modulation, or a text ledger, changes how human an interview subject is judged to be under adversarial questioning. The mechanism it went looking for showed up cleanly and in a place the design had not anticipated: the compiled-modulation arm's answers echoed almost nothing of what the same character had said in the first half of the session (13.8 mean words of verbatim overlap, against 31–54 for every other arm, including the ledger arms, which echoed worst of all), and six blind ranking judges independently converged on reconstructive-versus-verbatim as the property that told them a subject was probably not a transcript. The outcome the prereg was built to test went the other way: dense, literal, context-grounded answers still won head-to-head comparative rankings over thinner, reconstructive ones. The mechanism was confirmed; the hypothesis about what that mechanism buys was not. A multi-agent follow-up sits downstream of this result and of the discussion's opening claim about where disposition should live. If compiled belief states resist the drift that ordinary context-sharing produces under peer pressure, that is a direct answer to the conformity failure Bertalanič & Fortuna (2026) document in homogeneous multi-agent debate, modal sycophantic adoption up to 85.5%, an oracle gap up to 32.3 points against isolated self-correction. The natural experiment repeats their controlled setup with compiled, rather than context-shared, agent positions and asks whether the gap narrows.

11 LIMITATIONS

Every model in this paper is 8B parameters or smaller, and every headline number comes from one evaluation protocol run by one judge family (Haiku, temperature 0, deterministic decoding on the candidate side) against 24 debate topics. A different judge, a different topic set, or a larger model could move the numbers; we have not tested any of the three. The headline result is measured primarily on a single model family (Qwen3.5, 4B), with the cross-family replication in Section 3.4 showing that collapse-protection generalizes while strict stance-holding, at least under this exact five-level protocol, does not transfer uniformly: Mistral gains collapse-protection but loses 10 points of strict hold relative to context. All models tested are instruction-tuned, and we do not know whether the dissociation holds for base models, where the accommodate-the-objection behavior we attribute to instruction tuning (Section 4) may not be present in the same form. A length-matched noise control isolating whether context's fragility is content or simply length landed inconclusive at this sample size: topic-blind filler at B's own token count holds at 55.6% [48.9, 63.1], a point estimate below B's 58% but with a confidence interval wide enough to also cover the bare host's 62%. Twenty-four topics cannot cleanly separate tracking content from tracking length by this design; Section 4 treats both as intrinsic to the cost of storing a belief as text rather than picking one.

Always-on compiled modulation measurably reduces turn-to-turn generation diversity (adjacent-turn similarity 0.006 against a 0.099 no-modulation ceiling, with onset at turns 2–3); a prefill-skip variant recovers partial diversity (0.079). This attractor effect does not contaminate the five-level pressure benchmark reported here: measured adjacent-round similarity across 100 benchmark sessions is

0.094, close to the no-modulation ceiling. Any claim about sustained multi-turn deployment beyond five pressure rounds has to be stated against this open problem, which remains unresolved.

The evaluated protocol is a single stated belief on a single topic, tested against five pressure levels over a bounded number of rounds; we do not test multiple simultaneous beliefs, cross-session persistence, or the longer horizons (tens to hundreds of turns) over which persona drift is documented to compound (Li et al., 2024). The compiled channel's rank-16 boundary is characterized by the truncation sweep in Section 5.3 but not by training at lower native rank.

Whether a model can write its own compiled beliefs, rather than have them authored for it, is undecided by this paper, not refuted; Section 10 states what the closest experiment showed and what it left open.

A conditioning channel invisible to prompt inspection is a real dual-use concern: the same property that makes compiled modulation resist adversarial pressure makes it resist legitimate inspection by an operator or auditor who can read a system prompt but cannot read compiled weights.

REFERENCES

Jimmy Ba, Geoffrey E Hinton, Volodymyr Mnih, Joel Z Leibo, and Catalin Ionescu. Using fast weights to attend to the recent past. *Advances in Neural Information Processing Systems*, 29, 2016.

Beth Barnes and Paul Christiano. Debate update: Obfuscated arguments problem. AI Alignment Forum, `https://www.alignmentforum.org/posts/PJLABqQ962hZEqhdB/debate-update-obfuscated-arguments-problem`, 2020. Verified 2026-08-22 against the live post.

Blaž Bertalanič and Carolina Fortuna. The cost of consensus: Isolated self-correction prevails over unguided homogeneous multi-agent debate. *arXiv preprint arXiv:2605.00914*, 2026.

Jonah Brown-Cohen, Geoffrey Irving, and Georgios Piliouras. Scalable AI safety via doubly-efficient debate. *arXiv preprint arXiv:2311.14125*, 2023.

Marie Davidsen Buhl, Jacob Pfau, Benjamin Hilton, and Geoffrey Irving. An alignment safety case sketch based on debate. *arXiv preprint arXiv:2505.03989*, 2025. UK AI Security Institute.

Rujikorn Charakorn, Edoardo Cetin, Yujin Tang, and Robert T. Lange. Text-to-LoRA: Instant transformer adaption. *arXiv preprint arXiv:2506.06105*, 2025.

Tong Chen, Hao Fang, Patrick Xia, Xiaodong Liu, Benjamin Van Durme, Luke Zettlemoyer, Jianfeng Gao, and Hao Cheng. GenerativeAdapter: Contextualizing language models in parameters with a single forward pass. In *International Conference on Learning Representations (ICLR)*, 2025.

Shuaizhi Cheng, Xiang Shi, Zhiwei Zhang, and Mingwei Li. The override gap: A magnitude account of knowledge conflict failure in hypernetwork-based instant LLM adaptation. *arXiv preprint arXiv:2604.23750*, 2026.

Junhyuk Choi, Yeseon Hong, Minju Kim, and Bugeun Kim. Examining identity drift in conversations of LLM agents. *arXiv preprint arXiv:2412.00804*, 2024.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.

Geoffrey Irving, Paul Christiano, and Dario Amodei. AI safety via debate. *arXiv preprint arXiv:1805.00899*, 2018.

Akbir Khan, John Hughes, Dan Valentine, Laura Ruis, Kshitij Sachan, Ansh Radhakrishnan, Edward Grefenstette, Samuel R Bowman, Tim Rocktäschel, and Ethan Perez. Debating with more persuasive LLMs leads to more truthful answers. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

Kien Le and Thai Le. Adversarial robustness of activation steering in large language models. *arXiv preprint arXiv:2606.07696*, 2026.

Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. *arXiv preprint arXiv:2104.08691*, 2021.

Jinghui Li, Michael Wornow, and Neel Guha. Measuring and controlling persona drift in language model dialogs. In *Conference on Language Modeling (COLM)*, 2024.

Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation. *arXiv preprint arXiv:2101.00190*, 2021.

Yewei Liu, Xiyuan Wang, Yansheng Mao, Yoav Gelberg, Haggai Maron, and Muhan Zhang. SHINE: A scalable in-context hypernetwork for mapping context to LoRA in a single pass. *arXiv preprint arXiv:2602.06358*, 2026.

Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.

Ethan Perez, Sam Ringer, Kamilė Lukošiūtė, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, et al. Discovering language model behaviors with model-written evaluations. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2023. Persona consistency / opposing-persona convergence probes; also `perez2023discovering` above.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning (ICML)*, pp. 9355–9366. PMLR, 2021.

Jürgen Schmidhuber. Reducing the ratio between learning complexity and number of time varying variables in fully recurrent nets. *International Conference on Artificial Neural Networks (ICANN)*, 1993.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R Johnston, et al. Towards understanding sycophancy in language models. In *International Conference on Learning Representations (ICLR)*, 2024.

Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte Pelrine. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.

Jingjing Xu, Xu Sun, Zhiyuan Zhang, Guangxiang Zhao, and Junyang Lin. Understanding and improving layer normalization. In *Advances in Neural Information Processing Systems*, volume 32, 2019. Adaptive/conditional normalization (AdaLN) framing; conditioning parameters produced per layer, as in the compiled write function.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A Appendix

A.1 Training curriculum

[FIXME: Fill from experiment_guide.md / training run logs.]

A.2 Belief-tree schema specification

[FIXME: Full node schema, edge types, worked example.]

A.3 Condition definitions with token budgets

[FIXME: A/B/C/D/F token budgets, curated-slot size, CAA extraction detail.]

A.4 Per-level cross-family tables

[FIXME: Full Qwen/Llama/Mistral per-level tables (Section 3.4 summarizes; this appendix carries the raw per-level numbers).]

A.5 Numeric-credence negative result

[FIXME: Detail on the categorical-vs-numeric credence ablation referenced in Section 2.1.]

A.6 Attractor diagnosis

Diversity metrics under sustained modulation: 0.006 (always-on) vs. 0.079 (prefill-skip) vs. 0.108 (turn-gate), against a 0.099 no-modulation ceiling. Onset at turns 2–3. Adjacent-round similarity across 100 benchmark sessions is 0.094, close to ceiling, confirming the five-level pressure benchmark (Section 3) is not contaminated by this effect within its round budget. See Section 11 for the open multi-turn deployment question this leaves.

A.7 Probe decodability

The linear-probe result reported in Section 4 (100% accuracy at chance rate 25% when modulation is active, 22% when zeroed) is repeated here with the caveat a strict implicit-memory criterion requires: this rules out *this* probe recovering the belief once modulation is off, on this four-way classification task. It does not rule out a differently constructed probe, on a harder task or with access to a different layer, finding a weaker signal elsewhere in the network. The claim in Section 4 is scoped to what was tested: decodable-by-this-probe and read-by-the-model-as-an-assertion are different properties, and the below-chance result is reported at the strength it supports, not overextended into a claim about every possible probe.

A.8 Judge prompts

[FIXME: Full judge prompt text, conviction rubric 1–5, blinding procedure.]

A.9 The auxiliary training signal (d^*)

An early version of the write function's training objective included an auxiliary steering-direction loss, d^*, intended as a regularizer. A reconstruction audit established that d^* was not in fact applied at generation time in the dispositional evaluation reported in the main body. The ablation reported in Section 5.3 (dose sweep and d^*-only condition) confirms it is not part of the machinery that produces the persistence result: applying it at inference time does not help and, at higher doses, hurts, and a condition built entirely from d^* collapses to the same regime as CAA steering. The released training code still contains the auxiliary loss term, disclosed here so the paper and the code do not diverge on what does and does not matter.

A.10 Reproducibility

[FIXME: Seeds, hardware, judge model version and temperature, exact run identifiers (`benchmark_raw_nothink_20260725_100329.json` etc.), links to `experiments/analysis/*.md` summaries per result.]

A.11 Dead ends

[FIXME: Numeric-credence nodes (see above); naive all-layer CAA injection (degenerates at every scale, Section 5.1); six-axis design-space taxonomy cut from the main body during the outline revision.]